## Team Details

- ➔ Team name – NextGen Nexus
- ➔ Team leader name – Mohith Mandali
- ➔ Problem Statement - Conventional educational platforms lack adaptive flexibility required for neurodiverse students (e.g., ADHD, Dyslexia). There is a need for a real-time assistive tool that summarizes lectures into sensory-friendly formats, highlights key takeaways, and provides non-intrusive focus nudges..

## Brief about the idea

**NeuroLens AI** is a hyper-local, real-time AI assistant designed specifically for neurodiverse learners.

→It transforms live lectures into:

-Simplified language summaries
-Dyslexia-friendly structured notes
-Color-coded key takeaways
-Visual + audio reinforcement
-Gentle “focus nudges”

Unlike generic AI note-taking apps, NeuroLens is built on **Inclusion-by Design UX principles**, ensuring sensory safety, customizable contrast, and cognitive load reduction.

→It works both:

-In live classrooms
-During online lectures
- Offline via local device processing (for privacy)

# Opportunities:

- ➔ How different is it from any of the other existing ideas?
- ➔ How will it be able to solve the problem?
- ➔ USP of the proposed solution

→Unlike generic AI note-taking tools, NeuroLens AI is built specifically for neurodiverse cognition, not the average learner. It integrates real-time cognitive adaptation, sensory-friendly UI modes, and hyper-local processing for privacy and low latency. Instead of simply summarizing content, it dynamically restructures lectures into ADHD- and Dyslexia-friendly formats. The platform is designed with Inclusion-by-Design principles from the core architecture, not as an afterthought.

→ NeuroLens AI solves the problem by converting live lectures into simplified, structured, and visually optimized outputs that reduce cognitive overload. It highlights key takeaways instantly while filtering unnecessary noise. The system provides subtle focus nudges to help sustain attention without creating additional distraction. By allowing personalized contrast, font, and language modes, it ensures each learner can access content in the format that suits their cognitive style.

→ The USP of NeuroLens AI lies in its cognitive-aware adaptation engine that personalizes learning in real time. Its hyper-local AI processing ensures privacy, speed, and uninterrupted accessibility even in low-connectivity environments. The multi-sensory output system (visual, audio, structured summaries) enhances comprehension and retention. Most importantly, it operationalizes Inclusion-by-Design as a core product philosophy, making accessibility seamless rather than optional.

# List of features offered by the solution :

1 **Real-Time Lecture Transformation:** Live speech-to-text conversion with intelligent noise filtering
Instant extraction of key concepts and core arguments
Multi-level summarization (Quick | Standard | Deep)
Automatic highlighting of exam-relevant points

2 **Cognitive-Adaptive Learning Engine:** AI-driven attention pattern detection
Dynamic restructuring of content to reduce cognitive overload
Adjustable information density control
Personalized learning profiles for each student

3 **Inclusion-by-Design User Experience:** High-contrast, low-stimulation, and pastel calm themes
Dyslexia-friendly font options and optimized spacing
Simplified language mode (complex → clear & concise)
Minimal-motion interface to reduce sensory triggers

4 **Smart Focus Support System:** Non-intrusive visual focus nudges
Gentle audio cues (optional)
Micro-break recommendations based on engagement patterns
Real-time focus meter with adaptive prompts

5 **Multi-Sensory Output Modes:** Structured bullet summaries
Mind-map visualization mode
Audio reinforcement playback
Color-coded concept clustering

6 **Hyper-Local AI Processing**: On-device AI inference for ultra-low latency.
Works even in low or no internet conditions.
No raw lecture data stored externally.
Privacy-first encrypted data architecture.

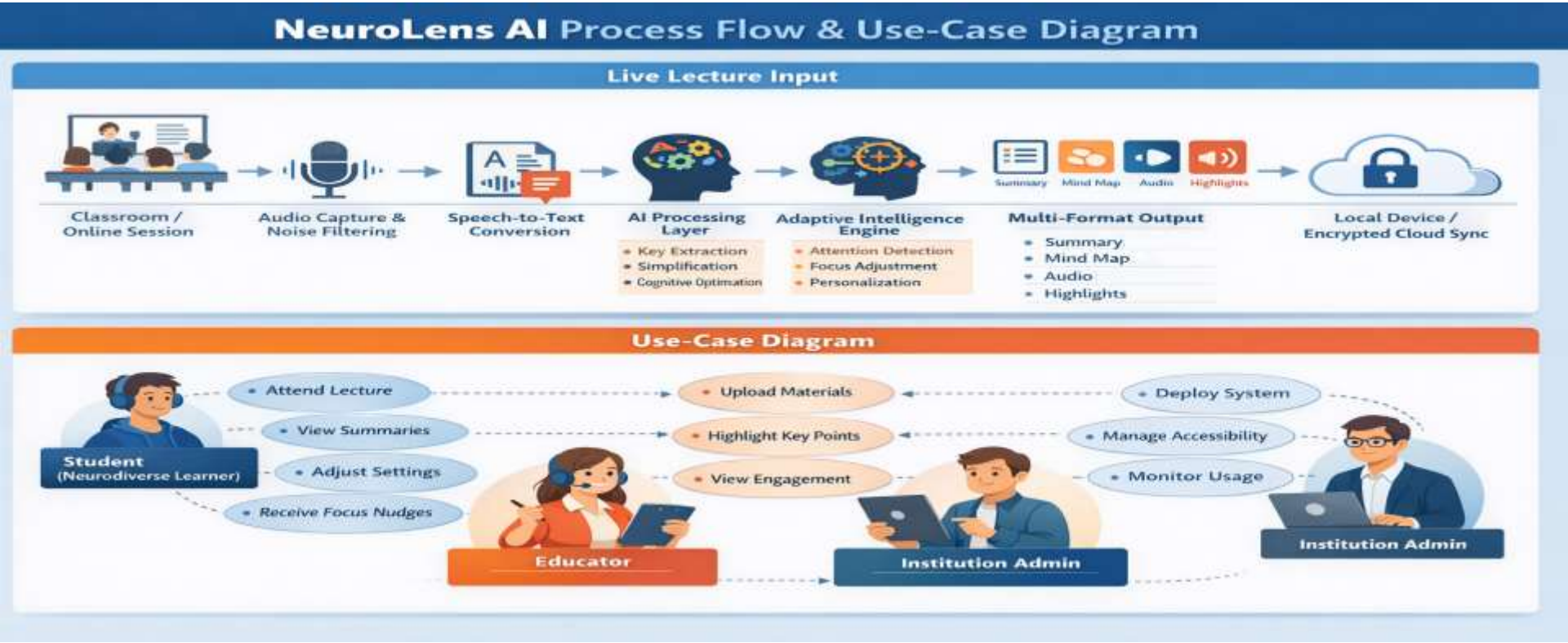
7 **Accessibility & Compliance Layer**: WCAG-aligned accessibility standards.
Adjustable text size, spacing, and background contrast.
Screen-reader compatibility.
Keyboard-only navigation support.

8 **Educator Integration Tools**: Instructor dashboard for summary preview.
Custom key-point emphasis control.
Class-level engagement analytics (privacy-safe).
Exportable structured notes.

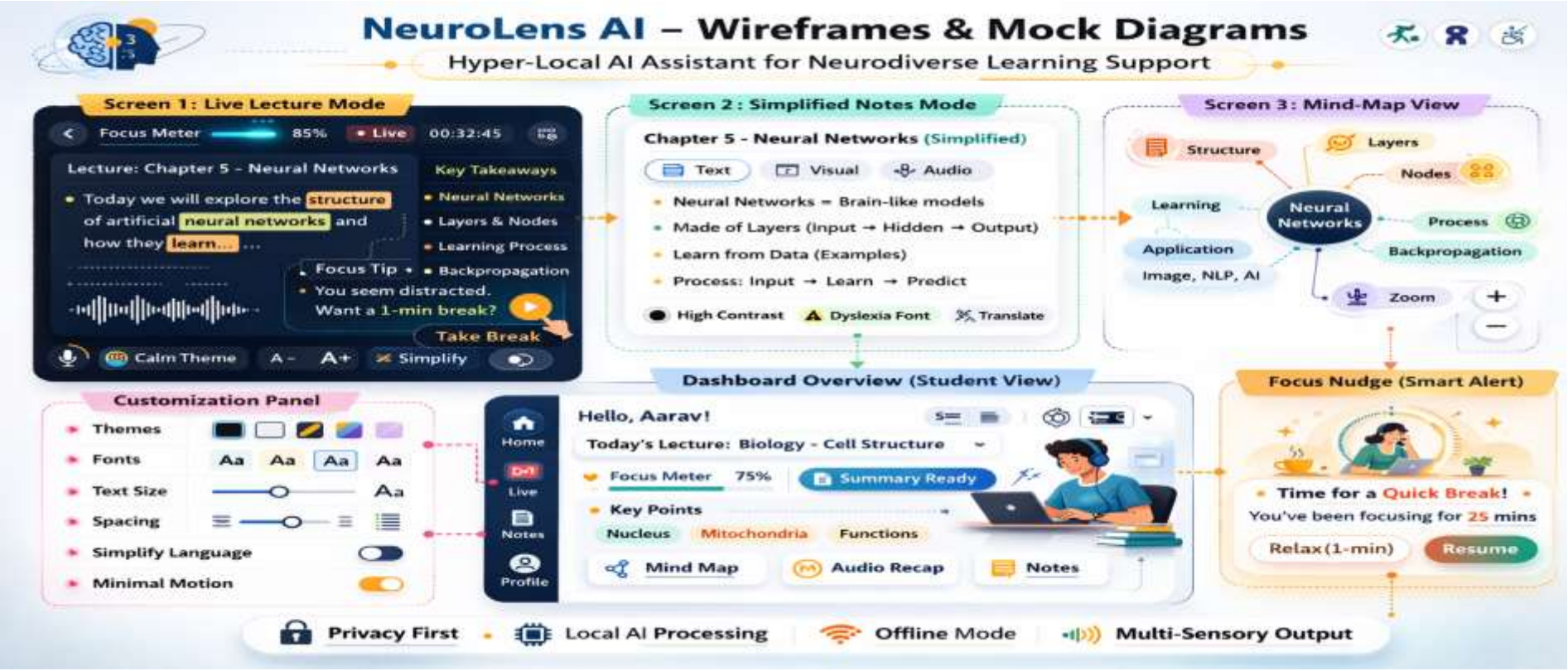
9 **Academic Performance Enhancers**: Smart revision sheets auto-generated after class.
Topic-wise recap mode.
Flashcard generation from lectures.
AI-powered question prediction based on patterns.

10 **Scalability & Institutional Readiness**: Modular architecture for school/college deployment.
LMS integration capability.
Secure academic data management.
Cloud-sync (optional) for cross-device access.

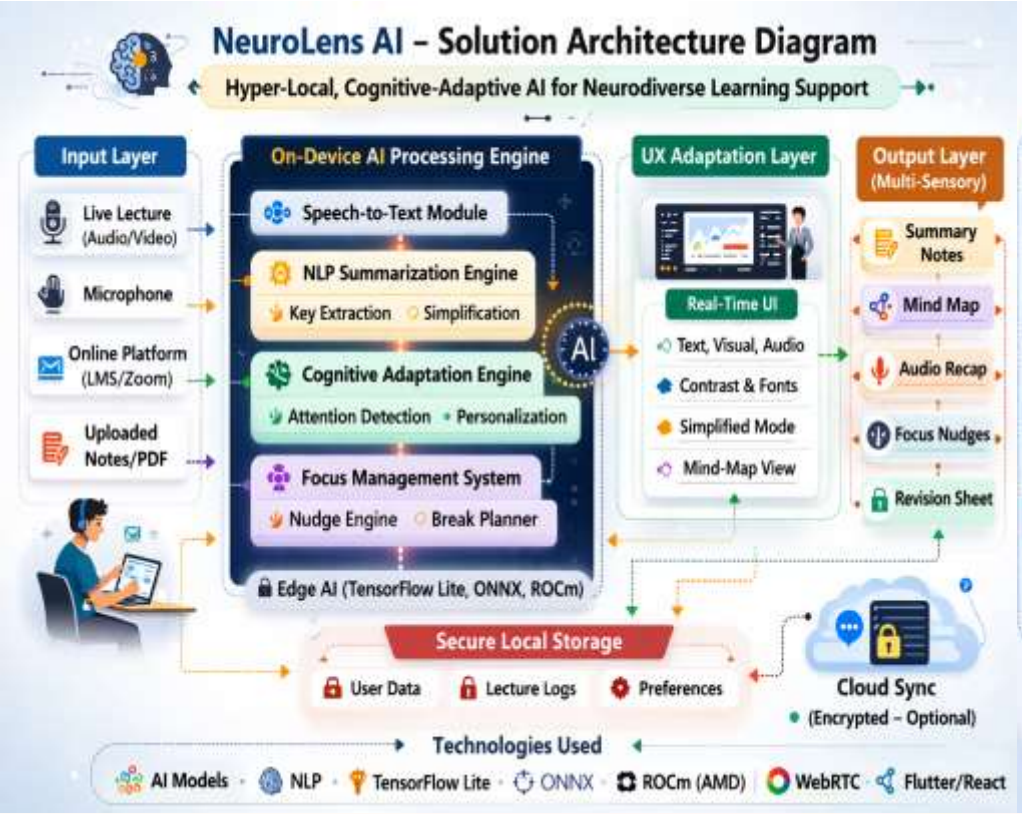
# Process flow diagram or Use-case diagram:

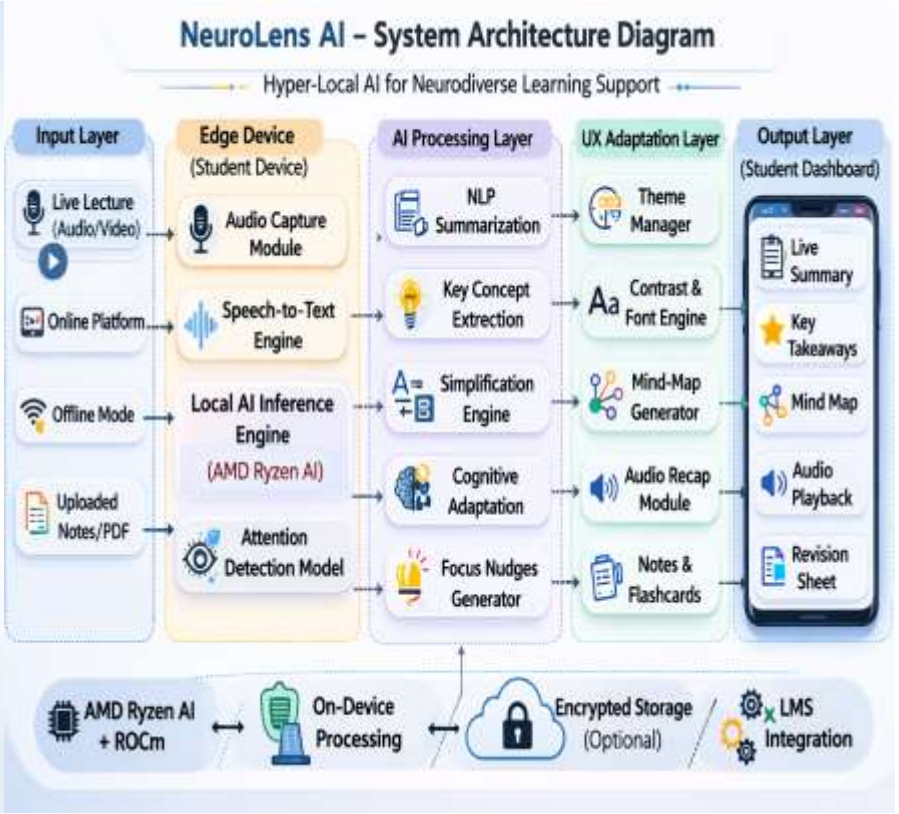


# Wireframes/Mock diagrams of the proposed solution (optional)



# Architecture diagram of the proposed solution



# Technologies to be used in the solution:

**1 Frontend – Flutter (Cross-Platform Accessible UI)**
Flutter enables a highly customizable, responsive, and inclusion-first interface across mobile, tablet, and web. It supports dynamic contrast themes, dyslexia-friendly fonts, minimal-motion mode, and real-time UI updates essential for neurodiverse learners.

**2 Backend – Supabase (Open-Source Backend-as-a-Service)**
Supabase provides secure authentication, real-time database (PostgreSQL), encrypted storage, and scalable APIs. It enables safe user data handling, preference storage, session tracking, and optional cloud sync — all with privacy-first architecture.

**3 AI & NLP – Transformer Models + ONNX Runtime (Edge AI)**
Transformer-based NLP models power intelligent summarization, key concept extraction, and simplified language conversion. ONNX Runtime enables optimized on-device inference for low latency, ensuring hyper-local AI processing without constant cloud dependency.

**4 Speech-to-Text – Advanced ASR Engine (Whisper / DeepSpeech-based)**
Automatic Speech Recognition converts live lectures into accurate transcripts with classroom noise filtering. This forms the foundation for real-time summarization and cognitive adaptation.

**5 Live Online Sessions – WebRTC (Low-Latency Streaming Protocol)**
WebRTC enables seamless real-time audio streaming for hybrid and online classrooms. It ensures smooth integration with LMS platforms while maintaining minimal delay for instant AI processing.

# Usage of AMD Products/Solutions:

**1. AMD Ryzen AI Accelerated Processors**

NeuroLens AI leverages AMD **Ryzen AI** processors for high-performance on-device machine learning inference. These processors deliver powerful neural network execution without relying on cloud resources — ensuring real-time summaries and personalization with minimal latency and maximum privacy.

**2. AMD Radeon Graphics for UI Rendering**

AMD **Radeon GPU** integrated graphics enable smooth, high-quality UI rendering and visualizations across Flutter-based interfaces. This enhances responsiveness and accessibility features (contrast themes, mind-map views), especially on tablets and laptops used in classrooms.

**3. AMD ROCm (Radeon Open Compute) Software Platform**

For advanced edge-AI training and optimization workflows, AMD **ROCm** provides an open, scalable compute platform. It accelerates model conversion and optimized runtime for transformer-based NLP engines used in lecture summarization and cognitive pattern detection.

**4. AMD Infinity Architecture for Low-Power AI**

The project leverages AMD's **Infinity Architecture** for power-efficient AI workloads on edge devices. This boosts battery life on mobile learning devices, making NeuroLens AI practical for all-day classroom and remote learning scenarios.

**5. Support for AMD Edge AI Ecosystem**

NeuroLens AI benefits from AMD's ecosystem tools for edge deployment, enabling optimized performance tuning, hardware-accelerated inferencing, and seamless integration with future AMD AI accelerators — ensuring longevity and scalability.

# Add as per the requirements of the contest:

🎯 **Innovation & Originality**
NeuroLens AI introduces a hyper-local, cognitive-adaptive learning assistant specifically engineered for neurodiverse students. Unlike conventional EdTech platforms, it integrates real-time AI summarization, inclusion-first UX, and non-intrusive focus management within a single unified system. The solution is original in its combination of Edge AI, accessibility design, and cognitive adaptation.

🌍 **Social Impact & Inclusivity**
The solution directly addresses accessibility gaps in mainstream education by supporting students with ADHD, Dyslexia, and sensory sensitivities. It promotes equitable learning opportunities by transforming traditional lectures into personalized, sensory-friendly formats. NeuroLens AI aligns with global inclusive education initiatives and accessibility standards.

⚙️ **Technical Feasibility**
The system is built using production-ready technologies including Flutter, Supabase, WebRTC, Transformer-based NLP, and Edge AI frameworks. AMD-powered on-device inference ensures low latency and privacy. The architecture is modular and scalable, enabling phased deployment across institutions.

📈 **Scalability & Market Potential**
NeuroLens AI can be deployed in schools, universities, coaching institutes, and corporate training environments. With LMS integration capability and hybrid compatibility (online + offline), the solution is ready for large-scale adoption. The global demand for inclusive EdTech solutions presents strong commercialization potential.

🔐 **Privacy & Ethical AI**
The platform follows a privacy-first architecture with local AI inference and encrypted storage. No raw lecture data is permanently stored without consent. The system ensures transparency, explainable summaries, and bias-aware AI processing.

🚀 **Future Roadmap**
Phase 1: Functional prototype with live summarization and adaptive UI
Phase 2: Institutional pilot deployment and educator analytics dashboard
Phase 3: Multilingual expansion and advanced cognitive personalization
Phase 4: Integration with national digital learning platforms

AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by H2S

# Thank you!